

Case 3 (Shortest Path)

G-Q: Find the shortest path between two nodes in an undirected graph. In an undirected graph, (i,j,k) means that node i and node j are connected with an undirected edge with weight k. Given a graph and a pair of nodes, you need to output the shortest path between the two nodes. Q: The nodes are numbered from 0 to 8, and the edges are: (0,1,4) (1,2,7) (1,7,1) (1,3,4) (2,6,2) (2,4,8) (2,7,5) (3,6,1) (4,8,3) (5,6,6) (6,8,8) (7,8,7). **Give the weight of the shortest path from node 0 to node 8.**

GPT-4: The weight of the shortest path from node 0 to node 8 is 14. ✕

GraphWiz: All the paths from node 0 to node 8 are:\n0,1,7,8 with a total weight of $\langle\langle 4 + 1 + 7 = 12 \rangle\rangle$, 0,1,3,6,8 with a total weight of $\langle\langle 4 + 4 + 1 + 8 = 17 \rangle\rangle$, 0,1,2,6,8 with a total weight of $\langle\langle 4 + 7 + 2 + 8 = 21 \rangle\rangle$, 0,1,2,4,8 with a total weight of $\langle\langle 4 + 7 + 8 + 3 = 22 \rangle\rangle$, 0,1,2,7,8 with a total weight of $\langle\langle 4 + 7 + 5 + 7 = 23 \rangle\rangle$. The weight of path 0,1,7,8 is the smallest, so the shortest path from node 0 to node 8 is [0,1,7,8] with a total weight of 12. ✓